MIXED DISCRETE AND CONTINUOUS DISTRIBUTION LIMITS FOR SHAPE PARAMETERS OF PURE QUARTIC NUMBER FIELDS

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Allowed standard prerequisites (editorial).

- Minkowski field shapes,integral bases of pure quartic fields,Lipschitz lattice counts and squarefree sieves. Original full family/type/parameter and measure definitions,all source-local lattice/sieve proofs with complete author bibliography retained.

1. Introduction

1.1. **Motivation and historical context.** The shape of a number field is a subtle arithmetic invariant which arises from the geometry of numbers. Given an inner product space $V \simeq \mathbb{R}^r$, the *shape* of a full rank lattice $\Lambda \subset V$ is an invariant which is unchanged by rotation, reflection and positive scalar dilations. Formally, the space of all such shapes is given by the double coset space

$$\mathcal{S}_r := \mathrm{GL}_r(\mathbb{Z}) \backslash \mathrm{GL}_r(\mathbb{R}) / \mathrm{GO}_r(\mathbb{R}),$$

where $\mathrm{GO}_r(\mathbb{R}) \subset \mathrm{GL}_r(\mathbb{R})$ denotes the group of matrices that preserve the standard inner product up to a positive scalar multiple. Let K be a number field of degree n and ring of integers $\mathcal{O}_K$. One can embed K into its Minkowski space

$$j : K \to K \otimes_{\mathbb{Q}} \mathbb{R} \simeq \mathbb{R}^n$$

and thus associate a lattice $j(\mathcal{O}_K) \subset \mathbb{R}^n$ to K. Here, $K \otimes_{\mathbb{Q}} \mathbb{R}$ is endowed with a natural inner product which arises from the trace form on K. The *shape* K is defined to be the shape of orthogonal projection of $j(\mathcal{O}_K)$ onto the hyperplane perpendicular to the vector $j(1)$.

All quadratic number fields have the same shape since all rank 1 lattices are dilations of a fixed lattice. For cubic number fields K, the associated rank 2 lattice is represented by a point in the complex upper half plane $\mathbb{H}$. The shape of K is represented by a point in the fundamental domain

$$\mathcal{F}_2 = \{x + iy \in \mathbb{H} \mid x \in [0, 1/2), \quad x^2 + y^2 \geq 1\},$$

which comes equipped with the hyperbolic measure, which is induced from the Haar measure on $\mathrm{GL}_2(\mathbb{R})$. D. Terr first introduced the study of the shape of a number field in his PhD thesis [Ter97], written under the supervision of H. Lenstra. He proved that the shapes of cubic fields are equidistributed in the space of shapes of rank-2 lattices. Furthermore, he showed that the shape associated to any Galois cubic field is always hexagonal. M. Bhargava and A. Shnidman [BS14] classified the shapes of cubic orders whose automorphism group is isomorphic to $\mathbb{Z}/3\mathbb{Z}$, and also established asymptotics for the number of such number fields. In a related development, G. Mantilla-Soler and M. Monsurró [MSM16] determined the shapes of cyclic number fields of prime degree ℓ.

M. Bhargava and Piper H. extended Terr's result to S_n number fields of degree $n = 4, 5$ and conjectured that such equidistribution results hold for all $n \geq 6$ as well. In greater detail, let μ denote the measure on $\mathcal{S}_{n-1}$ induced from the Haar measure on $\mathrm{GL}_{n-1}(\mathbb{R})$. An n-ic field is a number field $K/\mathbb{Q}$ of degree n. Given an n-ic field K, denote by $\widetilde{K}$ its Galois closure over $\mathbb{Q}$ and set

$$\mathrm{Gal}(K/\mathbb{Q}) := \mathrm{Gal}(\widetilde{K}/\mathbb{Q}).$$

We note that $\mathrm{Gal}(K/\mathbb{Q})$ acts faithfully on the n-embeddings $K \hookrightarrow \mathbb{C}$ and may thus be viewed as a subgroup of S_n. For $n = 3, 4, 5$ and a positive real number X, let $N_n^{(i)}(X)$ be the number of n-ic extensions of $\mathbb{Q}$ with i pairs of complex embeddings such that $\mathrm{Gal}(K/\mathbb{Q}) \simeq S_n$ and the absolute discriminant of K is less than X. Given a measurable set Y in $\mathcal{S}_{n-1}$ whose boundary has measure zero, let $N_n^{(i)}(X, Y)$ be the number of n-ic

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extensions of $\mathbb{Q}$ with i pairs of complex embeddings such that $\mathrm{Gal}(K/\mathbb{Q}) \simeq S_n$ and the absolute discriminant of K is less than X, and whose ring of integers has shape contained in Y. Then the main result in *loc. cit.* establishes that

$$\lim_{X\to\infty} \frac{N_n^{(i)}(X,Y)}{N_n^{(i)}(X)} = \frac{\mu(Y)}{\mu(\mathcal{S}_{n-1})}.$$

The proof of this result relies on the existence of parameterizations of cubic, quartic and quintic orders by Delone, Faddeev [DF64] and Bhargava [Bha04, Bha08]. Such parameterizations are not known in cases when $n \geq 6$.

In [BH16], the authors do mention that: "*It is an interesting problem to determine the distribution of lattice shapes for n-ic number fields having a given non-generic (i.e., non-S_n) associated Galois group, even heuristically.*" It is a natural question to study the distribution of shapes of families of non-S_n number fields $K/\mathbb{Q}$ of degree n. One natural family to consider is the family of *pure number fields* of degree n, namely, number fields of the form $K = \mathbb{Q}(m^{1/n})$, where m is an n-power free non-zero integer. Alternatively, this is the family of number fields $K/\mathbb{Q}$ of degree n, with Galois group $\mathrm{Gal}(\widetilde{K}/\mathbb{Q}) \simeq (\mathbb{Z}/n\mathbb{Z})^\times \ltimes (\mathbb{Z}/n\mathbb{Z})$ and having resolvent field $\mathbb{Q}(\mu_n)$. This question has been studied for pure cubic fields by R. Harron [Har17], and for pure prime degree number fields by E. Holmes [Hol22].

Let m be a positive cubefree integer and $\sqrt[3]{m}$ be the real third root of m and set $K_m := \mathbb{Q}(\sqrt[3]{m})$. These fields naturally fall into two families, distinguished by whether the prime (3) is tamely or wildly ramified in K_m. All other primes are either unramified or tamely ramified. Write $m = ab^2$, where a and b are coprime, squarefree, positive integers. Note that $K_m = \mathbb{Q}(\sqrt[3]{m}) = \mathbb{Q}(\sqrt[3]{m'})$, where $m' := a^2b$, since $mm' = (ab)^3$ is a cube. Assume without loss of generality, that $a > b$. Define the *ratio* of K_m to be $r_m := a/b$. Say that K_m is of *Type* I if $m \not\equiv \pm 1 \pmod 9$, and of *Type* II otherwise. Note that K_m is tamely ramified (or unramified) at (3) if and only if it is of Type I. Let α and β be the roots in K of $x^3 - m$ and $x^3 - m'$, respectively, so that $\beta = \alpha^2/b, \alpha = \beta^2/a$, and $K = \mathbb{Q}(\alpha) = \mathbb{Q}(\beta)$. The discriminant of K is $-3^3a^2b^2$ (resp. $-3a^2b^2$) for Type I (resp. Type II). For $m \equiv \pm 1 (\mathrm{mod} 9)$, the element $\nu = \left(1 \pm \alpha + \alpha^2\right)/3$ is in $\mathcal{O}_{K_m}$. An integral basis of K_m is given by:

$$\mathcal{O}_{K_m} = \begin{cases} \langle 1, \alpha, \beta\rangle & \text{if } K_m \text{ is of Type I,} \\ \langle 1, \nu, \beta\rangle & \text{if } K_m \text{ is of Type II,} \end{cases}$$

see [Har17, Lemma 2.1].

Theorem 1.1 (Theorem A [Har17])**.** *The shapes of pure cubic fields fall into two distinct loci:*

- *If K is of Type* I, its shape lies on the imaginary axis within $\mathcal{F}$, and is given explicitly by $ir_K^{1/3} \in \mathcal{F}$.
- If K is of Type II, its shape lies on the vertical line $\mathrm{Re}(z) = 1/3$, above height $1/3$; more precisely, the shape is $\frac{1+ir_K^{1/3}}{3} \in \mathbb{H}$.

To study the distribution of these shapes, one introduces measures on the sets $\mathscr{S}_{\mathrm{I}} := \{iy : y \geq 1\}$ and $\mathscr{S}_{\mathrm{II}} := \left\{\frac{1+iy}{3} : y \geq 1\right\}$. A natural choice is the measure induced by the hyperbolic metric on $\mathbb{H}$, or equivalently, the unique (up to scaling) measure invariant under the action of a subgroup of $\mathrm{SL}_2(\mathbb{R})$. Since the hyperbolic line element on $\mathbb{H}$ is

$$ds = \frac{\sqrt{dx^2 + dy^2}}{y},$$

this induces the measure $\frac{dy}{y}$ on both $\mathscr{S}_{\mathrm{I}}$ and $\mathscr{S}_{\mathrm{II}}$.

Alternatively, observe that the diagonal torus

$$\left\{\begin{pmatrix} y & 0 \\ 0 & y^{-1} \end{pmatrix}\right\} \subseteq \mathrm{SL}_2(\mathbb{R}),$$

acting on the base point $i \in \mathbb{H}$, defines a homeomorphism onto the positive imaginary axis, sending the subset with $y > 1$ onto $\mathscr{S}_{\mathrm{I}}$. This identifies $\mathscr{S}_{\mathrm{I}}$ with $(1, \infty)$ via $y \mapsto iy$, and thus endows it with the measure $\frac{dy}{y}$ induced from Haar measure on the torus. Conjugating the torus by an element of $\mathrm{SL}_2(\mathbb{R})$ yields the same

construction for $\mathscr{S}_{\mathrm{II}}$, giving rise to an invariant measure there as well. Denote the resulting measures on $\mathscr{S}_{\mathrm{I}}$ and $\mathscr{S}_{\mathrm{II}}$ by μ_{I} and μ_{II}, respectively. For $* \in \{\mathrm{I}, \mathrm{II}\}$, set

$$[R_1, R_2)_* = \begin{cases} i[R_1, R_2) & \text{if } * = \mathrm{I}, \\ \left(\frac{1+i[R_1,R_2)}{3}\right) & \text{if } * = \mathrm{II}. \end{cases}$$

Define

$$c_{\mathrm{I}} = \frac{2c\sqrt{3}}{15} \quad \text{and} \quad c_{\mathrm{II}} = \frac{c\sqrt{3}}{10}$$

where

$$c := \prod_p \left(1 - \frac{3}{p^2} + \frac{2}{p^3}\right)$$

the product being over all primes p.

Theorem 1.2 (Theorem C [Har17])**.** *Then, for all R_1, R_2,*

$$\lim_{X\to\infty} \frac{\#\left\{K \text{ of type } * \mid |\Delta_K| \le X, \operatorname{sh}_K \in [R_1, R_2)_*\right\}}{c_* \sqrt{X}} = \int_{[R_1,R_2)_*} d\mu_*$$

where $\Delta(K)$ is the discriminant of K and $\operatorname{sh}(K)$ is the shape of K (taken in $\mathscr{S}_*$).

More recently, Harron [Har21] has proven distribution results for complex quadratic cubic fields with fixed quadratic resolvent. In this case, the shape lies on a geodesic of the modular surface defined by the field's trace zero norm. Moreover, the shapes are equi-distributed with respect to the hyperbolic measure.

1.2. **Main results.** In this article, we study the distribution of shapes of the rings of integers of pure quartic number fields $K_m := \mathbb{Q}(\sqrt[4]{m})$, where m is a non-zero fourth power free and $m \neq -4$. For this family of number fields, the integral basis and discriminant were computed by Funakura [Fun84], and the classification gives five families, see Theorem 3.1. Suppose that m has sign $\pm$, then we say that m is of:

(1) Type I$\pm$: if $m \equiv 1 \pmod 8$,
(2) Type II$\pm$: if $m \equiv 2, 3 \pmod 4$,
(3) Type III$\pm$: if $m \equiv 4 \pmod{16}$ or $m \equiv 5 \pmod 8$,
(4) Type IV$\pm$: if $m \equiv 12 \pmod{32}$,
(5) Type V$\pm$: if $m \equiv 28 \pmod{32}$.

We write $m = ab^2c^3$ where a, b, c are squarefree and coprime integers. Moreover, assume that $|a| \ge c$ and that $a < 0$ if and only if m is negative. For each type, we associate an explicit matrix:

$$C_{\mathrm{I}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{ab}{4} & \frac{1}{4} & \frac{b}{4} & \frac{1}{4} \end{pmatrix}, \quad C_{\mathrm{II}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad C_{\mathrm{III}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix},$$

$$C_{\mathrm{IV}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad C_{\mathrm{V}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{b}{8} & \frac{1}{4} \end{pmatrix}.$$

The shape of K_m is then given by the projection of the matrix

$$C_* \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \sqrt{c/|a|} & 0 & 0 \\ 0 & 0 & 1/b & 0 \\ 0 & 0 & 0 & \sqrt{|a|/c} \end{pmatrix} C_*^T$$

to the orthogonal complement of the vector $(1, 1, 1, 1)$, where C_* is the matrix corresponding to the type of K_m. It thus turns out that the shape lies on one of 10 torus orbits depending on the sign and Type of m.

In all cases, the shape is determined by the *shape parameters*

$$\lambda(m) = (\lambda_1(m), \lambda_2(m)) := \left(\sqrt{\frac{c}{|a|}}, \frac{1}{b}\right),$$

and thus determined by the pair $(\frac{|a|}{c}, b) \in [1,\infty)^2$. We note that unlike the shape parameters considered in [Har17] and [Hol22], the parameter $\lambda_2(m) = \frac{1}{b}$ takes on a discrete set of values with an accumulation point at 0.

Harron shows that the shape is a complete invariant for the family of pure cubic fields (cf. [Har17, Theorem B]). In particular, if $\mathbb{Q}(\sqrt[3]{m_1})$ and $\mathbb{Q}(\sqrt[3]{m_2})$ are pure cubic fields whose rings of integers have the same shape, then $\mathbb{Q}(\sqrt[3]{m_1}) = \mathbb{Q}(\sqrt[3]{m_2})$. A similar statement holds for pure quartic fields, with a small caveat.

Theorem 1.3. *Let m_1 and m_2 be fourth-power–free integers such that $\mathrm{sh}(K_{m_1}) = \mathrm{sh}(K_{m_2})$. Then exactly one of the following holds:*

(1) $m_1 = -m_2$ and $m_1 \equiv m_2 \equiv 2 \pmod 4$,
(2) $m_1 = m_2$.

Our main result shows that there is a measure that gives the distribution of shapes, which is in fact the product of a discrete measure and a continuous measure. Define

$$\psi(x) := \sum_{n=1}^{\lfloor x \rfloor} \alpha(n), \quad \text{where} \quad \alpha(n) := \begin{cases} n^{-2/3} \prod_{\ell \mid n} \left(\frac{\ell}{\ell+2}\right) & \text{if } n \text{ is squarefree,} \\ 0 & \text{otherwise,} \end{cases}$$

and the product is over prime divisors ℓ of n. Let $\mu_\alpha(x)dx := \sum_n \alpha(n)\delta_n(x)dx$ be the sum of Dirac measures supported only at squarefree positive integers. Let $\hat{\mu}$ denote the measure $\frac{dx_1}{x_1}\mu_\alpha(x_2)dx_2$ on $[1,\infty)^2$ and define $\psi_*(x) := \sum_{n=1}^{\lfloor x \rfloor} M_*(n)\alpha(n)$. Below is the main result of the article.

Theorem 1.4. *Let $* \in \{\mathrm{I}, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{V}\}$ and let $\mathcal{R} := [R_1', R_1) \times [R_2', R_2) \subset [1,\infty)^2$. Then, we have that*

$$\begin{aligned} &\lim_{X\to\infty} \left(\frac{\#\left\{K_m \mid \pm m > 0,\ |\Delta_m| \le X,\ K_m \text{ is of Type } *,\ (|a|/c, b) \in \mathcal{R}\right\}}{2^{r_*/3} X^{1/3}} \right) \\ =&\frac{1}{2}\prod_\ell \left(1 - \frac{3}{\ell^2} + \frac{2}{\ell^3}\right) (\log R_1 - \log R_1') \left(\psi_*(R_2) - \psi_*(R_2')\right) \\ =&\frac{1}{2}\prod_\ell \left(1 - \frac{3}{\ell^2} + \frac{2}{\ell^3}\right) \int_{\mathcal{R}} M_*(\lfloor x \rfloor)\, d\hat{\mu}(x), \end{aligned}$$

where

$$\begin{aligned} M_{\mathrm{I}}(b) &:= \frac{\#\{(\bar{a},\bar{c}) \in (\mathbb{Z}/8\mathbb{Z})^2 \mid \bar{a}\bar{b}^2\bar{c}^3 = 1 \pmod 8\}}{8^2}, \\ M_{\mathrm{II}}(b) &:= \frac{\#\{(\bar{a},\bar{c}) \in (\mathbb{Z}/4\mathbb{Z})^2 \mid \bar{a}\bar{b}^2\bar{c}^3 = 2,3 \pmod 4\}}{16}, \\ M_{\mathrm{III}}(b) &:= \frac{\#\{(\bar{a},\bar{c}) \in (\mathbb{Z}/16\mathbb{Z})^2 \mid \bar{a}\bar{b}^2\bar{c}^3 = 4,5,10 \pmod{16}\}}{16^2}, \\ M_{\mathrm{IV}}(b) &:= \frac{\#\{(\bar{a},\bar{c}) \in (\mathbb{Z}/32\mathbb{Z})^2 \mid \bar{a}\bar{b}^2\bar{c}^3 = 12 \pmod{32}\}}{32^2}, \\ M_{\mathrm{V}}(b) &:= \frac{\#\{(\bar{a},\bar{c}) \in (\mathbb{Z}/32\mathbb{Z})^2 \mid \bar{a}\bar{b}^2\bar{c}^3 = 28 \pmod{32}\}}{32^2}, \end{aligned}$$

and

$$r_* = \begin{cases} 2 & \text{if } * = \mathrm{I} \text{ or } \mathrm{V}, \\ 4 & \text{if } * = \mathrm{III} \text{ or } \mathrm{IV}, \\ 8 & \text{if } * = \mathrm{II}. \end{cases}$$

1.3. **Organization.** Including the introduction, this article is divided into four sections. In §2, we review foundational concepts related to the shape of a number field. In §3, we compute the Gram matrices associated to pure quartic fields, using the explicit integral bases constructed in [Fun84]. This leads to a complete classification of the shapes of pure quartic fields into ten distinct classes. We give a proof of Theorem 1.3 at the end of this section. The main distribution results are established in §4. We begin by revisiting the notion of shape parameters and the Principle of Lipschitz. These, together with sieve methods, are then employed to estimate the number of integer lattice points in suitable regions. The section concludes with the proof of Theorem 1.4.

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2. Shapes of number fields

This section introduces notation and reviews background material on the shapes of number fields. Fix an integer $r \geq 2$, and let $M_r(\mathbb{R})$ denote the space of real $r \times r$ matrices. The group $\mathrm{GL}_r(\mathbb{R})$ consists of all invertible elements of $M_r(\mathbb{R})$, and denote by I_r the identity matrix. For any matrix $M \in M_r(\mathbb{R})$, write M^T for its transpose. Let $V \simeq \mathbb{R}^r$ be an inner product space with standard orthonormal basis $\{e_1, \ldots, e_r\}$, so that

$$(e_i, e_j) = \delta_{i,j} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Let $\mathrm{GO}_r(\mathbb{R})$ be the subgroup of $\mathrm{GL}_r(\mathbb{R})$ consisting of matrices M satisfying $M^T M = \lambda \mathrm{I}_r$ for some scalar $\lambda \in \mathbb{R}_{>0}$; these are precisely the invertible linear transformations preserving the inner product up to a positive scaling.

Definition 2.1. *Let $\Lambda \subset V$ be a lattice. The* shape *of Λ, denoted sh_Λ, is its equivalence class under the action of rotations, reflections, and positive scalar dilations. The space of all such shapes is the double coset space*

$$\mathcal{S}_r := \mathrm{GL}_r(\mathbb{Z}) \backslash \mathrm{GL}_r(\mathbb{R}) / \mathrm{GO}_r(\mathbb{R}).$$

Given a basis $\{b_1, \ldots, b_r\}$ for Λ, we express each b_i in terms of the standard basis as $b_i = \sum_{j=1}^r a_{i,j} e_j$. The matrix $(a_{i,j}) \in \mathrm{GL}_r(\mathbb{R})$ then determines a point in $\mathcal{S}_r$, namely the shape of Λ. This space is equipped with a natural measure $\mu_{\mathcal{S}_r}$ induced from Haar measure on $\mathrm{GL}_r(\mathbb{R})$.

Given any measurable subset $Y \subset \mathcal{S}_r$, we write $\mu(Y) := \mu_{\mathcal{S}_r}(Y)$ for its measure.

It is often convenient to describe $\mathcal{S}_r$ as the quotient of a discrete group acting on a homogenous space $\mathcal{H}_r$. Let $\mathcal{P}_r$ denote the space of positive definite $r \times r$ symmetric real matrices. The group $\mathrm{GL}_r(\mathbb{Z})$ acts on $\mathcal{P}_r$ by $G \mapsto M^T G M$, while $\lambda \in \mathbb{R}^\times$ acts by scaling, $G \mapsto \lambda^2 G$. Set $\mathcal{H}_r := \mathcal{P}_r / \mathbb{R}^\times$ and note that $\mathcal{H}_r$ inherits a natural $\mathrm{GL}_r(\mathbb{Z})$-action. We identify $\mathcal{H}_r$ with a subset of affine space $\mathbb{A}^{r^2} = \mathrm{M}_r(\mathbb{R})$ as follows:

$$\mathcal{H}_r = \{M \in \mathrm{M}_r(\mathbb{R}) \mid M \text{ is positive definite and symmetric with } \det M = 1\}.$$

There is a natural bijection $\mathrm{GL}_r(\mathbb{R}) / \mathrm{GO}_r(\mathbb{R}) \xrightarrow{\sim} \mathcal{H}_r$ given by $M \mapsto |\det M|^{-2/r} M^T M$, inducing an identification $\mathcal{S}_r \simeq \mathrm{GL}_r(\mathbb{Z}) \backslash \mathcal{H}_r$. This identification can be made explicit as follows. If Λ is a lattice with basis $\{b_1, \ldots, b_r\}$, its Gram matrix $\mathrm{Gr}(\Lambda)$ is defined by $(\langle b_i, b_j \rangle)_{i,j}$, and the shape of Λ corresponds to the class of $\mathrm{Gr}(\Lambda)$ in $\mathcal{S}_r$. This class is invariant under change of basis by $M \in \mathrm{GL}_r(\mathbb{Z})$ and rescaling by $\lambda \in \mathbb{R}^\times$, as is seen from the following relation

$$\mathrm{Gr}(\lambda M \cdot \Lambda) = \lambda^2 M^T \mathrm{Gr}(\Lambda) M,$$

demonstrating that sh_Λ is indeed well defined.

The description of $\mathcal{H}_2$ is quite simple, since it consists of matrices $\begin{pmatrix} a & b \\ b & c \end{pmatrix} \in \mathrm{M}_2(\mathbb{R})$ with $a > 0$ and $ac - b^2 = 1$. Let $\mathbb{H}$ denote the upper half plane in $\mathbb{C}$. Let $\mathrm{GL}_2^+(\mathbb{R})$ be the subgroup of $\mathrm{GL}_2(\mathbb{R})$ consisting of invertible matrices with positive determinant. There is a natural action of $\mathrm{GL}_2(\mathbb{R})$ on $\mathbb{H}$ by fractional linear transformations:

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}(z) := \left(\frac{\alpha z + \beta}{\gamma z + \delta}\right).$$

This action is transitive and the stabilizer of i consists of matrices in $\mathrm{GO}_2^+(\mathbb{R})$. This gives an identification

$$\mathrm{GL}_2(\mathbb{R})/\,\mathrm{GO}_2(\mathbb{R}) \xrightarrow{\sim} \mathbb{H}$$

induced by the map $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \mapsto \left(\frac{\alpha z+\beta}{\gamma z+\delta}\right)$. Consider the matrix

$$M = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix},$$

where $y > 0$ and let

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} = MM^T = \begin{pmatrix} (y + y^{-1}x^2) & y^{-1}x \\ y^{-1}x & y^{-1} \end{pmatrix}.$$

There is a $\mathrm{GL}_2(\mathbb{R})$-equivariant identification $\mathcal{H}_2 \xrightarrow{\sim} \mathbb{H}$ sending $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ to $z = x + iy \in \mathbb{H}$, where $x = \frac{b}{c}$ and $y = \sqrt{\frac{a}{c} - x^2}$. A fundamental domain for the action of $\mathrm{GL}_2(\mathbb{Z})$ on $\mathbb{H}$ is then given by

$$\mathcal{F}_2 = \{x + iy \mid x \in [0, 1/2), \quad y > 0, \quad x^2 + y^2 \geq 1\}.$$

For $r > 2$, describing a fundamental domain for the action of $\mathrm{GL}_r(\mathbb{Z})$ on $\mathcal{H}_r$ becomes significantly more intricate. We focus here on the case $r = 3$, and give an explicit description of a fundamental domain for the action of $\mathrm{GL}_3(\mathbb{Z})$ on $\mathcal{H}_3$. Since $\mathrm{SL}_3(\mathbb{Z})$ is a subgroup of index two in $\mathrm{GL}_3(\mathbb{Z})$, with the two cosets distinguished by determinant ± 1, it suffices to describe a fundamental domain for the action of $\mathrm{SL}_3(\mathbb{Z})$ on $\mathcal{H}_3$.

The resulting fundamental domain $\mathcal{F}_3$ can be viewed as a higher-rank generalization of the classical fundamental domain for the $\mathrm{SL}_2(\mathbb{Z})$-action on the upper half-plane. To realize $\mathcal{H}_3$ as an analogue of an upper half-space, we adopt Iwasawa coordinates, expressing each point $Y \in \mathcal{H}_3$ as

$$Y = \begin{pmatrix} 1 & x_1 & x_2 \\ 0 & 1 & x_3 \\ 0 & 0 & 1 \end{pmatrix}^T \begin{pmatrix} y_1 & 0 & 0 \\ 0 & y_2 & 0 \\ 0 & 0 & y_3 \end{pmatrix} \begin{pmatrix} 1 & x_1 & x_2 \\ 0 & 1 & x_3 \\ 0 & 0 & 1 \end{pmatrix},$$

where $y_j > 0$, $x_j \in \mathbb{R}$, and $\prod_{j=1}^3 y_j = 1$.

A fundamental domain for the action of $\mathrm{SL}_3(\mathbb{Z})$ on $\mathcal{H}_3$, due to Minkowski, is given by the region

$$\mathcal{M}_3 = \left\{ Y = (y_{ij}) \in \mathcal{H}_3 \;\middle|\; \begin{array}{ll} y_{11} \leq y_{22} \leq y_{33}, & 0 \leq y_{12} \leq \frac{1}{2} y_{11}, \\ 0 \leq y_{23} \leq \frac{1}{2} y_{22}, & |y_{13}| \leq \frac{1}{2} y_{11}, \\ e^t Y e \geq y_{33} & \text{for all } e = (\pm 1, \pm 1, \pm 1) \end{array} \right\}.$$

It follows that

$$0 \leqslant x_1 \leqslant \frac{1}{2}, \quad |x_2| \leqslant \frac{1}{2}, \quad 0 \leqslant y_1 x_1 x_2 + y_2 x_3 \leqslant \frac{1}{2}\left(y_1 x_1^2 + y_2\right), \tag{2.1}$$

and

$$-\frac{1}{4} y_1/y_2 \leqslant x_3 \leqslant \frac{1}{2} + (3/8) y_1/y_2. \tag{2.2}$$

Hence, we get

$$\mathcal{F}_3 = \{v = (x_1, x_2, x_3, y_1, y_2) \in \mathbb{R}^5 \quad | \quad v \text{ satisfies (2.1) and (2.2) }\}.$$

Now we discuss the shape of a number field K with $[K : \mathbb{Q}] = n$ and ring of integers $\mathcal{O}_K$. Let $\{1, \alpha_1, \ldots, \alpha_{n-1}\}$ be an integral basis. Let $\{\sigma_i\}_{1 \le i \le n}$ denote the set of embeddings of K into $\mathbb{C}$. The Minkowski embedding

$$\begin{aligned} j : K &\to \mathbb{C}^n \\ \alpha &\mapsto \big(\sigma_1(\alpha), \ldots, \sigma_n(\alpha)\big) \end{aligned}$$

identifies K with an n-dimensional subspace of $\mathbb{C}^n$. The real span of the image, denoted $K_{\mathbb{R}}$, inherits a natural inner-product structure, and the restriction of j to $\mathcal{O}_K$ yields a rank-n lattice in $K_{\mathbb{R}}$. Although one might be inclined to define the shape of K as the shape of this lattice, such a definition presents difficulties when studying the distribution of shapes across a family of number fields. This is because each lattice necessarily contains the vector $j(1)$, this introduces a nontrivial constraint, disrupting randomness in the distribution of shapes. To circumvent this issue, we instead define the shape of K in terms of a sublattice obtained by projecting $j(\mathcal{O}_K)$ onto the orthogonal complement of $j(1)$. This is accomplished via the map

$$\alpha^{\perp} := n\alpha - \mathrm{tr}(\alpha),$$

which sends elements of $\mathcal{O}_K$ to those of trace zero. The set $\mathcal{O}_K^{\perp}$, consisting of the images of $\mathcal{O}_K$ under this map, forms a rank $r := (n-1)$ submodule of $\mathcal{O}_K$.

Definition 2.2. *We define the shape of K, denoted sh_K, to be the shape of the lattice $j(\mathcal{O}_K^{\perp})$ in $\mathcal{S}_{n-1}$.*

Given a basis $\{1, \alpha_1, \ldots, \alpha_{n-1}\}$ of $\mathcal{O}_K$, the corresponding basis for $\mathcal{O}_K^{\perp}$ is given by $\{\alpha_1^{\perp}, \ldots, \alpha_{n-1}^{\perp}\}$. As we discussed above, the Gram matrix is given as follows:

$$\mathrm{Gr}\left(j\left(\mathcal{O}_K^{\perp}\right)\right) = \big(\langle j(\alpha_i^{\perp}), j(\alpha_j^{\perp})\rangle\big). \tag{2.3}$$

3. Integral basis and shapes of pure quartic number fields

Let $m \in \mathbb{Z}$ be fourth-power free, and let β_m be a root of $X^4 - m \in \mathbb{Q}[X]$, with $\arg(\beta_m) = 0$ if $m > 0$, and $\arg(\beta_m) = \pi/4$ if $m < 0$. The field $K_m := \mathbb{Q}(\beta_m)$ is called a *pure quartic field*. Up to replacing m by m^3, we assume in this section that $m = abc^3$, where:

(i) $a \neq 1$, and a, b, c are pairwise coprime, squarefree integers;
(ii) $b, c > 0$;
(iii) $|a| \ge c$ if a is odd;
(iv) c is odd;
(v) $m \neq -4$.

Condition (v) excludes the reducible case $X^4 + 4$. The assumptions (iii) and (iv) are in place in order to compute an integral basis and thus describe the shape of K_m. Since isomorphic number fields have the same shape, we shall in the next section replace (i)-(v) above with the following:

(i') $a \neq 1$, and a, b, c are pairwise coprime, squarefree integers;
(ii') $b, c > 0$;
(iii') $|a| \ge c$;
(iv') $m \neq -4$.

We henceforth assume m is in this normal form, and define

$$\alpha = \frac{\beta_m^2}{bc}, \quad \beta = \beta_m, \quad \gamma = \frac{\beta_m^3}{bc^2}.$$

An explicit integral basis for $K_m = \mathbb{Q}(\beta_m)$ was computed by Funakura [Fun84], as recalled below.

Theorem 3.1 (Funakura). *Given an integer m as above, $\mathcal{B} = \{1, \lambda, \mu, \nu\}$ given below forms an integral basis of K_m:*

m	1	λ	μ	ν
1(mod8)	1	$\frac{1+\alpha}{2}$	β	$\frac{ab+\alpha+b\beta+\gamma}{4}$
2(mod4) 3(mod4)	1	α	β	γ
4(mod16) 5(mod8)	1	$\frac{1+\alpha}{2}$	β	$\frac{\beta+\gamma}{2}$
12(mod32)	1	α	$\frac{1+\alpha+\beta}{2}$	$\frac{\beta+\gamma}{2}$
28(mod32)	1	α	$\frac{1+\alpha+\beta}{2}$	$\frac{4\alpha+b\beta+2\gamma}{8}$.

Proof. See [Fun84, Theorem 1]. □

We shall denote the integral basis by $\mathcal{B}_*$ when K_m is of Type $*$. As a corollary to the above result, Funakura computes the discriminant Δ_m of K_m.

Corollary 3.2 (Funakura). *With respect to notation above, we have that*

$$\Delta_m = \begin{cases} -2^2a^3b^2c^3 & \text{if } m \equiv 1 \mod 8, \quad 28 \mod 32, \\ -2^4a^3b^2c^3 & \text{if } m \equiv 4 \mod 16, \quad 5 \mod 8, \quad 12 \mod 32, \\ -2^8a^3b^2c^3 & \text{if } m \equiv 2, 3 \mod 4. \end{cases}$$

Proof. The above result is [Fun84, Corollary 1]. □

Theorem 3.1 provides an explicit integral basis in all cases under consideration, since m is not divisible by 8. For each such case, we now determine the shape of the corresponding lattice K_m by computing the Gram matrix. The table above suggests that the cases $m \equiv 2, 3 \pmod 4$ admit particularly simple bases. We begin by computing the Gram matrix in this setting and subsequently use change-of-basis matrices to treat the remaining cases.

If $m > 0$, K_m has two real embeddings. Let σ_1 (resp. σ_2) be the embedding $K \hookrightarrow \mathbb{C}$ sending β to β (resp. $-\beta$). Let τ be the complex embedding mapping $\beta \mapsto i\beta$. On the other hand, when $m < 0$, let τ_1 (resp. τ_2) be the embedding defined by $\beta \mapsto i\beta$ (resp. $\beta \mapsto -\beta$). Let j_1 and j_2 denote the embeddings of K into $\mathbb{C}^4$ corresponding to the cases $m > 0$ and $m < 0$, respectively. These are defined by

$$j_1(a) = \big(\sigma_1(a), \sigma_2(a), \tau(a), \overline{\tau(a)}\big), \quad \text{and} \quad j_2(a) = \big(\tau_1(a), \tau_2(a), \overline{\tau_2(a)}, \overline{\tau_1(a)}\big).$$

Let $\mathcal{P}_{*,\pm}$ denote the Gram matrix when m is of Type $*$ and sign $\pm$.

Proposition 3.3. *The Gram matrix of the basis $\mathcal{B}_{\text{II}} = \{1, \alpha, \beta, \gamma\}$ is given by :*

$$\mathcal{P}_{\text{II},+} = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4\alpha^2 & 0 & 0 \\ 0 & 0 & 4\beta^2 & 0 \\ 0 & 0 & 0 & 4\gamma^2 \end{pmatrix}, \quad \mathcal{P}_{\text{II},-} = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4|\alpha|^2 & 0 & 0 \\ 0 & 0 & 4|\beta|^2 & 0 \\ 0 & 0 & 0 & 4|\gamma|^2 \end{pmatrix}.$$

Proof. Let $m > 0$. Then we have:

(1) $j_1(1) = (1, 1, 1, 1)$,
(2) $j_1(\alpha) = (\alpha, \alpha, -\alpha, -\alpha)$,
(3) $j_1(\beta) = (\beta, -\beta, i\beta, -i\beta)$,
(4) $j_1(\gamma) = (\gamma, -\gamma, -i\gamma, i\gamma)$.

Taking the standard Hermitian inner product on $\mathbb{C}^4$, we compute:

$$\langle j_1(\lambda), j_1(\lambda) \rangle = 4\lambda^2$$

for each $\lambda \in \{1, \alpha, \beta, \gamma\}$, and

$$\langle j_1(\lambda_1), j_1(\lambda_2) \rangle = 0$$

for all distinct $\lambda_1, \lambda_2 \in \{1, \alpha, \beta, \gamma\}$. That is, the vectors $j_1(1), j_1(\alpha), j_1(\beta), j_1(\gamma)$ form an orthogonal set, and the corresponding Gram matrix is diagonal with entries $4\lambda^2$ on the diagonal, as claimed.

Now suppose $m < 0$. Then:

(1) $j_2(1) = (1, 1, 1, 1)$,
(2) $j_2(\beta) = (\beta, i\beta, -i\bar{\beta}, \bar{\beta})$,
(3) $j_2(\alpha) = \left(\frac{\beta^2}{bc}, -\frac{\beta^2}{bc}, -\frac{\bar{\beta}^2}{bc}, \frac{\bar{\beta}^2}{bc}\right)$,
(4) $j_2(\gamma) = \left(\frac{\beta^3}{bc^2}, i\frac{\beta^3}{bc^2}, -i\frac{\bar{\beta}^3}{bc^2}, \frac{\bar{\beta}^3}{bc^2}\right)$.

It is easy to see that the vectors $j_2(\alpha), j_2(\beta), j_2(\gamma)$ are mutually orthogonal with respect to the standard Hermitian inner product. The squared norm of $j_2(\alpha)$ is:

$$\begin{aligned}\langle j_2(\alpha), j_2(\alpha)\rangle &= \left\langle \left(\frac{\beta^2}{bc}, -\frac{\beta^2}{bc}, -\frac{\bar{\beta}^2}{bc}, \frac{\bar{\beta}^2}{bc}\right), \left(\frac{\beta^2}{bc}, -\frac{\beta^2}{bc}, -\frac{\bar{\beta}^2}{bc}, \frac{\bar{\beta}^2}{bc}\right)\right\rangle \\ &= \frac{1}{(bc)^2}\left(|\beta^2|^2 + |\beta^2|^2 + |\bar{\beta}^2|^2 + |\bar{\beta}^2|^2\right) = 4\left(\frac{|\beta|^2}{bc}\right)^2 = 4|\alpha|^2.\end{aligned}$$

A similar computation gives:

$$\langle j_2(\beta), j_2(\beta)\rangle = 4|\beta|^2, \quad \langle j_2(\gamma), j_2(\gamma)\rangle = 4|\gamma|^2.$$

Hence, the Gram matrix of the set $\{j_2(1), j_2(\alpha), j_2(\beta), j_2(\gamma)\}$ is diagonal with entries $4|\lambda|^2$, and the vectors are pairwise orthogonal as desired.

□

Proposition 3.4. *The Gram matrix of the basis* $\mathcal{B}_{\mathrm{I}} = \{1, \frac{1+\alpha}{2}, \beta, \frac{ab+\alpha+b\beta+\gamma}{4}\}$ *is given by :*

$$\mathcal{P}_{\mathrm{I},+} = \begin{pmatrix} 4 & 2 & 0 & ab \\ 2 & 1+\alpha^2 & 0 & \frac{ab+\alpha^2}{2} \\ 0 & 0 & 4\beta^2 & b\bar{\beta}^2 \\ ab & \frac{ab+\alpha^2}{2} & b\beta^2 & \frac{ab^2+b\alpha^2+b^2\beta^2+\gamma^2}{4} \end{pmatrix}, \quad \mathcal{P}_{\mathrm{I},-} = \begin{pmatrix} 4 & 2 & 0 & ab \\ 2 & 1+|\alpha|^2 & 0 & \frac{ab+|\alpha|^2}{2} \\ 0 & 0 & 4|\beta|^2 & b|\beta|^2 \\ ab & \frac{ab+|\alpha|^2}{2} & b|\beta|^2 & \frac{ab^2+b|\alpha|^2+b^2|\beta|^2+|\gamma|^2}{4} \end{pmatrix}.$$

Proof. The change of basis matrix from $\mathcal{B}_{\mathrm{II}}$ to $\mathcal{B}_{\mathrm{I}}$ is given as

$$C_{\mathrm{II}} := \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ ab/4 & 1/4 & b/4 & 1/4 \end{pmatrix}.$$

Therefore,

$$\begin{aligned}\mathcal{P}_{\mathrm{I},+} = C_{\mathrm{II}}\,\mathcal{P}_{\mathrm{II},+}\,C_{\mathrm{II}}^T &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ ab/4 & 1/4 & b/4 & 1/4 \end{pmatrix}\begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4\alpha^2 & 0 & 0 \\ 0 & 0 & 4\beta^2 & 0 \\ 0 & 0 & 0 & 4\gamma^2 \end{pmatrix}\begin{pmatrix} 1 & 1/2 & 0 & ab/4 \\ 0 & 1/2 & 0 & 1/4 \\ 0 & 0 & 1 & b/4 \\ 0 & 0 & 0 & 1/4 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 2 & 0 & ab \\ 2 & 1+\alpha^2 & 0 & \frac{ab+\alpha^2}{2} \\ 0 & 0 & 4\beta^2 & b\bar{\beta}^2 \\ ab & \frac{ab+\alpha^2}{2} & b\beta^2 & \frac{ab^2+b\alpha^2+b^2\beta^2+\gamma^2}{4} \end{pmatrix}.\end{aligned}$$

Similarly,

$$\mathcal{P}_{\mathrm{I},-} = C_{\mathrm{II}}\,\mathcal{P}_{\mathrm{II},-}\,C_{\mathrm{II}}^T = \begin{pmatrix} 4 & 2 & 0 & ab \\ 2 & 1+|\alpha|^2 & 0 & \frac{ab+|\alpha|^2}{2} \\ 0 & 0 & 4|\beta|^2 & b|\bar{\beta}|^2 \\ ab & \frac{ab+|\alpha|^2}{2} & b|\beta|^2 & \frac{ab^2+b|\alpha|^2+b^2|\beta|^2+|\gamma|^2}{4} \end{pmatrix}.$$

□

Proposition 3.5. *The Gram matrix of the basis* $\mathcal{B}_{\rm III} = \{1, \frac{1+\alpha}{2}, \beta, \frac{\beta+\gamma}{2}\}$ *is given by :*

$$\mathcal{P}_{\rm III,+} = \begin{pmatrix} 4 & 2 & 0 & 0 \\ 2 & 1+\alpha^2 & 0 & 0 \\ 0 & 0 & 4\beta^2 & 2\beta^2 \\ 0 & 0 & 2\beta^2 & \beta^2+\gamma^2 \end{pmatrix}, \quad \mathcal{P}_{\rm III,-} = \begin{pmatrix} 4 & 2 & 0 & 0 \\ 2 & 1+|\alpha|^2 & 0 & 0 \\ 0 & 0 & 4|\beta|^2 & 2|\beta|^2 \\ 0 & 0 & 2|\beta|^2 & |\beta|^2+|\gamma|^2 \end{pmatrix}$$

Proof. The change of basis matrix from $\mathcal{B}_{\rm II}$ to $\mathcal{B}_{\rm III}$ is given as

$$C_{\rm III} := \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}.$$

Therefore,

$$\mathcal{P}_{\rm III,+} = C_{\rm III}\, \mathcal{P}_{\rm II,+}\, C_{\rm III}^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4\alpha^2 & 0 & 0 \\ 0 & 0 & 4\beta^2 & 0 \\ 0 & 0 & 0 & 4\gamma^2 \end{pmatrix} \begin{pmatrix} 1 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 1/2 \end{pmatrix}$$
$$= \begin{pmatrix} 4 & 2 & 0 & 0 \\ 2 & 1+\alpha^2 & 0 & 0 \\ 0 & 0 & 4\beta^2 & 2\beta^2 \\ 0 & 0 & 2\beta^2 & \beta^2+\gamma^2 \end{pmatrix}.$$

A similar relation holds when m is negative. □

Proposition 3.6. *The Gram matrix of the basis* $\mathcal{B}_{\rm IV} = \{1, \alpha, \frac{1+\alpha+\beta}{2}, \frac{\beta+\gamma}{2}\}$ *is given by :*

$$\mathcal{P}_{\rm IV,+} = \begin{pmatrix} 4 & 0 & 2 & 0 \\ 0 & 4\alpha^2 & 2\alpha^2 & 0 \\ 2 & 2\alpha^2 & 1+\alpha^2+\beta^2 & \beta^2 \\ 0 & 0 & \beta^2 & \gamma^2 \end{pmatrix}, \quad \mathcal{P}_{\rm IV,-} = \begin{pmatrix} 4 & 0 & 2 & 0 \\ 0 & 4|\alpha|^2 & 2|\alpha|^2 & 0 \\ 2 & 2|\alpha|^2 & 1+|\alpha|^2+|\beta|^2 & |\beta|^2 \\ 0 & 0 & |\beta|^2 & |\gamma|^2 \end{pmatrix}.$$

Proof. The change of basis matrix from $\mathcal{B}_{\rm II}$ to $\mathcal{B}_{\rm IV}$ is given as

$$C_{\rm IV} := \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}.$$

Therefore,

$$\mathcal{P}_{\rm IV,+} = C_{\rm IV}\, \mathcal{P}_{\rm II,+}\, C_{\rm IV}^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4\alpha^2 & 0 & 0 \\ 0 & 0 & 4\beta^2 & 0 \\ 0 & 0 & 0 & 4\gamma^2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 0 & 1/2 \end{pmatrix}$$
$$= \begin{pmatrix} 4 & 0 & 2 & 0 \\ 0 & 4\alpha^2 & 2\alpha^2 & 0 \\ 2 & 2\alpha^2 & 1+\alpha^2+\beta^2 & \beta^2 \\ 0 & 0 & \beta^2 & \gamma^2 \end{pmatrix}.$$

A similar relation holds when m is negative. □

Proposition 3.7. *The Gram matrix of the basis* $\mathcal{B}_{\rm V} = \{1, \alpha, \frac{1+\alpha+\beta}{2}, \frac{4\alpha+b\beta+2\gamma}{8}\}$ *is given by :*

$$\mathcal{P}_{\rm V,+} = \begin{pmatrix} 4 & 0 & 2 & 0 \\ 0 & 4\alpha^2 & 2\alpha^2 & 2\alpha^2 \\ 2 & 2\alpha^2 & 1+\alpha^2+\beta^2 & \alpha^2+\frac{b}{4}\beta^2 \\ 0 & 2\alpha^2 & \alpha^2+\frac{b}{4}\beta^2 & \alpha^2+\frac{b^2}{64}\beta^2+\frac{1}{4}\gamma^2 \end{pmatrix}$$

and

$$\mathcal{P}_{\mathrm{V},-} = \begin{pmatrix} 4 & 0 & 2 & 0 \\ 0 & 4|\alpha|^2 & 2|\alpha|^2 & 2|\alpha|^2 \\ 2 & 2|\alpha|^2 & 1+|\alpha|^2+|\beta|^2 & |\alpha|^2+\frac{b}{4}|\beta|^2 \\ 0 & 2|\alpha|^2 & |\alpha|^2+\frac{b}{4}|\beta|^2 & |\alpha|^2+\frac{b^2}{64}|\beta|^2+\frac{1}{4}|\gamma|^2 \end{pmatrix}.$$

Proof. The change of basis matrix from $\mathcal{B}_{\mathrm{II}}$ to $\mathcal{B}_{\mathrm{V}}$ is given as

$$C_{\mathrm{V}} := \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 1/2 & 1/2 & 0 \\ 0 & 1/2 & b/8 & 1/4 \end{pmatrix}.$$

Therefore,

$$\begin{aligned} \mathcal{P}_{\mathrm{V},+} = C_{\mathrm{V}}\,\mathcal{P}_{\mathrm{II},+}\,C_{\mathrm{V}}^T &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 1/2 & 1/2 & 0 \\ 0 & 1/2 & b/8 & 1/4 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4\alpha^2 & 0 & 0 \\ 0 & 0 & 4\beta^2 & 0 \\ 0 & 0 & 0 & 4\gamma^2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & 1 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & b/8 \\ 0 & 0 & 0 & 1/4 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 0 & 2 & 0 \\ 0 & 4\alpha^2 & 2\alpha^2 & 2\alpha^2 \\ 2 & 2\alpha^2 & 1+\alpha^2+\beta^2 & \alpha^2+\frac{b}{4}\beta^2 \\ 0 & 2\alpha^2 & \alpha^2+\frac{b}{4}\beta^2 & \alpha^2+\frac{b^2}{64}\beta^2+\frac{1}{4}\gamma^2 \end{pmatrix}. \end{aligned}$$

A similar relation holds when m is negative. □

We summarize the results above.

Proposition 3.8. *When K_m is of Type $*$ and m has sign $\pm$, there is an invertible matrix C_* such that*

$$\mathcal{P}_{*,\pm} = C_* \mathcal{P}_{\mathrm{II},\pm} C_*^T.$$

Here, the matrix C_* is given by:

$$C_{\mathrm{I}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ ab/4 & 1/4 & b/4 & 1/4 \end{pmatrix},$$

$$C_{\mathrm{II}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$C_{\mathrm{III}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix},$$

$$C_{\mathrm{IV}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix},$$

$$C_{\mathrm{V}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 1/2 & 1/2 & 0 \\ 0 & 1/2 & b/8 & 1/4 \end{pmatrix}.$$

Proof. The result follows from Propositions 3.4, 3.5, 3.6 and 3.7 (and their proofs). □

Proof of Theorem 1.3. By Proposition 3.8, the shape of K_m lies in one of five distinct loci, determined by the Type of m. These sets are disjoint since distinct torus orbits in the space of shapes are disjoint. Thus if

$$\mathrm{sh}(K_{m_1}) = \mathrm{sh}(K_{m_2}),$$

then m_1 and m_2 must have the same type. Write

$$m_i = a_i b_i^2 c_i^3, \qquad i = 1, 2,$$

where a_i, b_i, c_i are pairwise coprime squarefree integers with $b_i > 0$ and $c_i > 0$. Equality of shapes implies

$$\lambda_1(m_1) = \lambda_1(m_2), \qquad \lambda_2(m_1) = \lambda_2(m_2),$$

and therefore

$$\frac{c_1}{|a_1|} = \frac{c_2}{|a_2|}, \qquad b_1 = b_2.$$

Thus $a_1 c_2 = \pm a_2 c_1$. Since $(a_i, c_i) = 1$ for $i = 1, 2$, it follows that

$$a_1 = \pm a_2, \qquad b_1 = b_2, \qquad c_1 = c_2.$$

If $a_1 = a_2$, then $m_1 = m_2$ and we are done. Suppose instead that $a_1 = -a_2$, so that $m_1 = -m_2$. Because m_1 and $m_2 = -m_1$ have the same Type (as their shapes coincide), the sign change does not move m_1 out of its Type. By inspection, this occurs only when $m_1 \equiv 2 \pmod 4$. This proves the theorem. □

4. Distribution results

For the sake of discussion, assume that $m > 0$ and $m \equiv 2, 3 \pmod 4$. The shape sh_m of $j(\mathcal{O}_{K_m})^\perp \in \mathcal{S}_3$, is given by

$$\mathrm{sh}_m = \begin{pmatrix} \alpha^2 & & \\ & \beta^2 & \\ & & \gamma^2 \end{pmatrix} = \begin{pmatrix} \beta^4/(bc)^2 & & \\ & \beta^2 & \\ & & \beta^6/(bc^2)^2 \end{pmatrix}.$$

After permuting the variables, we have that:

$$\mathrm{sh}_m = \begin{pmatrix} \gamma_1^2 & & \\ & \gamma_2^2 & \\ & & \gamma_3^2 \end{pmatrix}$$

where $\gamma_1 := \beta$, $\gamma_2 := \beta^2/bc$ and $\gamma_3 := \beta^3/bc^2$. We scale this matrix by dividing by (abc). Let $\lambda_i = \gamma_i^2/(abc)$, we find that $\mathrm{sh}_m = \mathrm{diag}\,(\lambda_1, \lambda_2, \lambda_3)$,

$$\lambda_1^2 = \frac{\beta^4}{(abc)^2} = \frac{ab^2c^3}{a^2b^2c^2} = \frac{c}{a},$$
$$\lambda_2 = \frac{\beta^4}{(bc)^2(abc)} = \frac{ab^2c^3}{ab^3c^3} = \frac{1}{b},$$
$$\lambda_3^2 = \frac{\beta^{12}}{(bc^2)^4(abc)^2} = \frac{a^3b^6c^9}{a^2b^6c^{10}} = \frac{a}{c}.$$

Since β is real and $a, b, c > 0$, we find that $\lambda_i > 0$ for $i = 1, 2, 3$, and therefore,

$$\lambda_1 = \sqrt{\frac{c}{a}}, \quad \lambda_2 = \frac{1}{b} \quad \text{and} \quad \lambda_3 = \sqrt{\frac{a}{c}}.$$

We find that

$$\mathrm{sh}_m = \begin{pmatrix} \sqrt{\frac{c}{a}} & & \\ & \frac{1}{b} & \\ & & \sqrt{\frac{a}{c}} \end{pmatrix},$$

we refer to the pair

$$\lambda(m) = (\lambda_1(m), \lambda_2(m)) = \left(\sqrt{\frac{c}{a}}, \frac{1}{b}\right) \tag{4.1}$$

as the *shape parameter*.

In the more general case, suppose that m is of Type $*$. Assume that the conditions (i')-(iv') are satisfied and set

$$\lambda(m) = (\lambda_1(m), \lambda_2(m)) = \left(\sqrt{\frac{c}{|a|}}, \frac{1}{b}\right).$$

The shape of K_m is then given by the projection of the matrix

$$C_* \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \sqrt{c/|a|} & 0 & 0 \\ 0 & 0 & 1/b & 0 \\ 0 & 0 & 0 & \sqrt{|a|/c} \end{pmatrix} C_*^T$$

to the orthogonal complement of the vector $(1, 1, 1, 1)$.

Let $\mathcal{R} \subset \mathbb{R}^2$ be a closed and bounded region whose boundary is rectifiable and has total length L. The result below is often attributed to Lipschitz, with an early reference appearing in the work of Bachmann [Bac04], who credits Lipschitz as the source (see also [Dav51]).

Lemma 4.1 (Lipschitz Principle)**.** *Let $\mathcal{R} \subset \mathbb{R}^2$ be as above. Then the number of integer lattice points contained in $\mathcal{R}$ satisfies*

$$\left|\#(\mathcal{R} \cap \mathbb{Z}^2) - \text{Area}(\mathcal{R})\right| \leq 4(L+1).$$

Consider the torus orbit $\mathcal{T} \subset \mathcal{H}_3$ consisting of diagonal matrices of the form

$$D(x_1, x_2) = x_2^{1/3} \begin{pmatrix} x_1^{-1} & & \\ & x_2^{-1} & \\ & & x_1 \end{pmatrix}.$$

Define

$$\psi(x) := \sum_{n=1}^{\lfloor x \rfloor} \alpha(n), \quad \text{where} \quad \alpha(n) := \begin{cases} n^{-2/3} \prod_{\ell \mid n} \left(\frac{\ell - 1}{\ell + 1}\right) & \text{if } n \text{ is squarefree,} \\ 0 & \text{otherwise,} \end{cases}$$

and the product is over prime divisors ℓ of n. Let $\hat{\mu}$ denote the measure $\frac{\alpha(\lfloor x_2 \rfloor)}{x_1}\, dx_1\, dx_2$ on $\mathcal{T}$.

Given a pair (R_1, R_2), define the region

$$W(R_1, R_2) := \left\{ D(x_1, x_2) \in \mathcal{T} \;\middle|\; x_1^2 \in [1, R_1],\ x_2 \in [1, R_2] \right\}.$$

Then we find that

$$\hat{\mu}\left(W(R_1, R_2)\right) = \int_1^{R_2} \int_1^{\sqrt{R_1}} \frac{\alpha(\lfloor x_2 \rfloor)}{x_1}\, dx_1\, dx_2 = \frac{1}{2} \log R_1 \cdot \psi(R_2).$$

Let $\mathcal{R}(N, R_1, R_2)$ be defined as follows:

$$\mathcal{R}(N, R_1, R_2) := \{(a, b, c) \in \mathbb{R}^3_{\geq 1} \mid ab^{2/3}c < N, \quad \frac{a}{c} \in [1, R_1] \quad \text{and} \quad b \in [1, R_2]\}.$$

Let $\mathcal{R}_{\mathbb{Z}^3}(N, R_1, R_2) := \mathcal{R}(N, R_1, R_2) \cap \mathbb{Z}^3$. For $M, R \geq 1$, define

$$\mathcal{S}(M, R) := \left\{ (a, c) \in \mathbb{R}^2_{\geq 1} \;\middle|\; ac < M,\ \frac{a}{c} \in [1, R] \right\},$$

and let $\mathcal{S}_{\mathbb{Z}^2}(M, R) := \mathcal{S}(M, R) \cap \mathbb{Z}^2$. Then we have the identity

$$\mathcal{R}_{\mathbb{Z}^3}(N, R_1, R_2) = \bigsqcup_{b=1}^{\lfloor R_2 \rfloor} \mathcal{S}_{\mathbb{Z}^2}\left(\frac{N}{b^{2/3}}, R_1\right), \tag{4.2}$$

where the union is disjoint.

Lemma 4.2. *The area of $\mathcal{S}(M, R)$ is given by:*

$$\text{Area}\left(\mathcal{S}(M, R)\right) = \begin{cases} \frac{M}{2} \log R - \frac{(R-1)}{2} & \text{if } M \geq R; \\ \frac{M}{2} \log M - \frac{(M-1)}{2} & \text{if } M < R. \end{cases}$$

Proof. Note that for any fixed $c \in [1, \sqrt{M}]$, the condition $ac < M$ and $\frac{a}{c} \in [1, R]$ together imply that

$$a \in \left[c, \min(Rc, M/c)\right].$$

We begin by analyzing the case $M \geq R$. Accordingly, we split the region of integration into two parts:

$$\begin{aligned}
\text{Area}(\mathcal{S}(M,R)) &= \int_1^{\sqrt{M/R}} \int_c^{Rc} da\, dc + \int_{\sqrt{M/R}}^{\sqrt{M}} \int_c^{M/c} da\, dc \\
&= \int_1^{\sqrt{M/R}} (R-1)c\, dc + \int_{\sqrt{M/R}}^{\sqrt{M}} \left(\frac{M}{c} - c\right) dc \\
&= \frac{R-1}{2}\left(\frac{M}{R}\right) - \frac{1}{2}\left(M - \frac{M}{R}\right) + \frac{M}{2}\log R \\
&= \frac{M}{2}\log R - \frac{R-1}{2}.
\end{aligned}$$

Next, suppose $M < R$. In this case, for all $c \in [1, \sqrt{M}]$, we have $M/c \leq Rc$, so the upper bound of the inner integral is simply M/c. Thus,

$$\begin{aligned}
\text{Area}(\mathcal{S}(M,R)) &= \int_1^{\sqrt{M}} \int_c^{M/c} da\, dc = \int_1^{\sqrt{M}} \left(\frac{M}{c} - c\right) dc \\
&= M\log\sqrt{M} - \frac{1}{2}(M-1) = \frac{M}{2}\log M - \frac{M-1}{2}.
\end{aligned}$$

□

Lemma 4.3. *Let $L = L(M,R)$ be the length of the boundary of $\mathcal{S}(M,R)$. Then, we have that*

$$L \leq \left(\sqrt{2} + \sqrt{1+R^2}\right)(\sqrt{M} - 1) + (R-1).$$

Proof. Assume that $M \geq R$. Then the total boundary length is given by $L = L_1 + L_2 + L_3 + L_4$, where the individual contributions are as follows:

- $L_1 = \text{length}\{(c,c) \mid c \in [1, \sqrt{M}]\} = \sqrt{2}(\sqrt{M} - 1)$,
- $L_2 = \text{length}\{(c, Rc) \mid c \in [1, \sqrt{M/R}]\} = \sqrt{1+R^2}(\sqrt{M/R} - 1)$,
- $L_3 = \text{length}\{(c, M/c) \mid c \in [\sqrt{M/R}, \sqrt{M}]\} = \int_{\sqrt{M/R}}^{\sqrt{M}} \sqrt{1 + \left(\frac{M}{c^2}\right)^2}\, dc \leq \sqrt{1+R^2}\left(\sqrt{M} - \sqrt{M/R}\right)$,
- $L_4 = R - 1$ is the length of the vertical segment from $(1,1)$ to $(1,R)$.

Thus, the total length is given by

$$L \leq \left(\sqrt{2} + \sqrt{1+R^2}\right)(\sqrt{M} - 1) + (R-1).$$

Similar analysis applies in the case when $M < R$ to give the result. □

Proposition 4.4. *We have that*

$$\#\mathcal{R}_{\mathbb{Z}^3}(N, R_1, R_2) = \frac{N}{2}\log R_1 \left(\sum_{n=1}^{\lfloor R_2 \rfloor} n^{-2/3}\right) + O_{R_1,R_2}(\sqrt{N}).$$

Proof. From Lemma 4.1 and Lemmas 4.2 and 4.3, it follows that

$$\left| \#\left(\mathcal{S}(M,R) \cap \mathbb{Z}^2\right) - \frac{M}{2}\log R \right| \leq 4(L+1) \leq CR\sqrt{M},$$

where $C > 0$ is an absolute constant. Noting that

$$\#\mathcal{R}_{\mathbb{Z}^3}(N, R_1, R_2) = \sum_{b=1}^{\lfloor R_2 \rfloor} \#\mathcal{S}_{\mathbb{Z}^2}(N/b^{2/3}, R_1),$$

we deduce that

$$\left| \#\mathcal{R}_{\mathbb{Z}^3}(N, R_1, R_2) - \frac{N}{2}\left(\sum_{b=1}^{\lfloor R_2 \rfloor} b^{-2/3}\right) \log R_1 \right| \leq C R_1 \sqrt{R_2}\sqrt{N}.$$

□

Remark 4.5. *Higher-dimensional analogues of Lipschitz's principle, due to Davenport* [Dav51]*, allow for effective volume estimates by bounding the lower-dimensional projections (or "shadows") of a region containing the lattice points; see, for example,* [Hol22, Corollary 6.13]*. However, in our situation, the area of the two-dimensional projection onto the (a, c)-plane is too large to yield meaningful results.*

Definition 4.6. *Let ℓ be a prime number. A triple $(a, b, c) \in \mathbb{Z}^3_{>0}$ is said to be ℓ-carefree if*

$$\ell^2 \nmid ab, \quad \ell^2 \nmid bc, \quad \ell^2 \nmid ca.$$

The triple (a, b, c) is said to be carefree if each of the products ab, bc, and ca is squarefree. Equivalently, (a, b, c) is carefree if and only if it is ℓ-carefree for all primes ℓ.

Given a positive integer n, we say that a triple $(a, b, c) \in \mathbb{Z}^3_{>0}$ is n-carefree if it is ℓ-carefree for all primes ℓ dividing n. Fix a residue class τ modulo 32 such that $8 \nmid \tau$. Let $\mathcal{C}^\tau(N, R_1, R_2)$ (respectively, $\mathcal{C}^\tau_n(N, R_1, R_2)$) denote the set of triples $(a, b, c) \in \mathcal{R}_{\mathbb{Z}^3}(N, R_1, R_2)$ that are carefree (respectively, n-carefree) and $ab^2c^3 \equiv \tau \pmod{32}$. Then

$$\mathcal{C}^\tau(N, R_1, R_2) = \bigcap_\ell \mathcal{C}^\tau_\ell(N, R_1, R_2),$$

where the intersection is taken over all prime numbers ℓ.

Let $n > 0$ be squarefree and odd. Define Ω_n to be the set of residue classes $(\bar{a}, \bar{b}, \bar{c}) \in (\mathbb{Z}/32n^2\mathbb{Z})^3$ such that for each prime ℓ dividing n, one has that $\ell^2 \nmid \bar{a}\bar{b}$, $\ell^2 \nmid \bar{b}\bar{c}$, $\ell^2 \nmid \bar{c}\bar{a}$, and moreover, $\bar{a}\bar{b}^2\bar{c}^3 \equiv \tau \pmod{32}$. Given an odd prime ℓ, let $\mathcal{A}_\ell$ denote $(\bar{a}, \bar{b}, \bar{c}) \in (\mathbb{Z}/\ell^2\mathbb{Z})^3$ such that $\ell^2 \nmid \bar{a}\bar{b}$, $\ell^2 \nmid \bar{b}\bar{c}$, $\ell^2 \nmid \bar{c}\bar{a}$. Denote by $\mathcal{A}^\tau_2$ the set of $(\bar{a}, \bar{b}, \bar{c}) \in (\mathbb{Z}/32\mathbb{Z})^3$ such that $4 \nmid \bar{a}$ and $\bar{a}\bar{b}^2\bar{c}^3 \equiv \tau \pmod{32}$. Clearly by the Chinese Remainder Theorem,

$$\Omega_n = \mathcal{A}^\tau_2 \times \prod_{\ell \mid n} \mathcal{A}_\ell,$$

where ℓ ranges over prime divisors of n. Define

$$\mathfrak{d}_\ell := \frac{\#\mathcal{A}_\ell}{\#(\mathbb{Z}/\ell^2\mathbb{Z})^3} = \frac{\#\mathcal{A}_\ell}{\ell^6},$$

if ℓ is an odd prime. On the other hand, set

$$\mathfrak{d}^\tau_2 := \frac{\#\mathcal{A}^\tau_2}{\#(\mathbb{Z}/32\mathbb{Z})^3} = \frac{\#\mathcal{A}^\tau_2}{32^3}.$$

For each squarefree natural number b, define $n_\tau(b)$ to be the number of pairs $(\bar{a}, \bar{c}) \in (\mathbb{Z}/32\mathbb{Z})^2$ such that $\tau = \bar{a}\bar{b}^2\bar{c}^3$.

Lemma 4.7. *For any odd prime ℓ, we have*

$$\#\mathcal{A}_\ell = (\ell^2 - \ell)^2\big(\ell^2 + 2\ell - 3\big), \qquad \textit{and hence} \qquad \mathfrak{d}_\ell = \left(1 - \frac{1}{\ell}\right)^2 \left(1 + \frac{2}{\ell} - \frac{3}{\ell^2}\right).$$

Proof. The number of tuples $(\bar{a}, \bar{b}, \bar{c}) \in (\mathbb{Z}/\ell^2\mathbb{Z})^3$ such that $\ell \nmid \bar{a}\bar{b}\bar{c}$ is $(\ell^2 - \ell)^3$, since each coordinate must be a unit modulo ℓ. The number of triples for which exactly one coordinate is divisible by ℓ is

$$3 \cdot \ell(\ell - 1)\,(\ell^2 - \ell)^2,$$

where the factor $\ell(\ell - 1)$ counts elements divisible by ℓ but not by ℓ^2 in $\mathbb{Z}/\ell^2\mathbb{Z}$, and the factor 3 accounts for the choice of the coordinate.

In total, the number of triples $(\bar{a}, \bar{b}, \bar{c})$ for which at most one coordinate is divisible by ℓ is

$$(\ell^2 - \ell)^3 \;+\; 3\ell(\ell - 1)(\ell^2 - \ell)^2 = (\ell^2 - \ell)^2\big(\ell^2 + 2\ell - 3\big).$$

Dividing by ℓ^6, we obtain

$$\mathfrak{d}_\ell = \frac{(\ell^2-\ell)^2(\ell^2+2\ell-3)}{\ell^6} = \left(1-\frac{1}{\ell}\right)^2\left(1+\frac{2}{\ell}-\frac{3}{\ell^2}\right).$$

□

Proposition 4.8. *Let n be an odd squarefree natural number which is divisible by all primes $\ell \leq R_2$ and τ be a residue class mod 32 such that $8 \nmid \tau$. Then for $N > n^4R_1R_2$,*

$$\#\mathcal{C}_n^\tau(N,R_1,R_2) = \frac{N}{2\times 32^2}\prod_{\ell\mid n}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1 \sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor} b^{-2/3}n_\tau(b)\prod_{\ell\mid b}\left(\frac{\ell}{\ell+2}\right) + O_{n,R_1,R_2}(\sqrt{N}),$$

where $O_{n,R_1,R_2}(\cdot)$ indicates that the constant in the error bound depends on R_1, R_2 and n.

Proof. Let Λ_n be the lattice $(32n^2\mathbb{Z})^3 \subset \mathbb{Z}^3$. Note that $\Omega_n \subset \mathbb{Z}^3/\Lambda_n$ choose a set of representatives $\widetilde{\Omega}_n \subset ([0,32n^2)\cap\mathbb{Z})^3$. We find that

$$\begin{aligned}\mathcal{C}_n^\tau(N,R_1,R_2) &= \bigsqcup_{z\in\widetilde{\Omega}_n}(z+\Lambda_n)\cap\mathcal{R}(N,R_1,R_2),\\ &= \bigsqcup_{z\in\widetilde{\Omega}_n}\Lambda_n\cap\left(\mathcal{R}(N,R_1,R_2)-z\right).\end{aligned}$$

Observe that

$$\#\left(\Lambda_n\cap\left(\mathcal{R}(N,R_1,R_2)-z\right)\right) = \#\left(\mathbb{Z}^3\cap\left(\frac{1}{32n^2}\cdot\mathcal{R}(N,R_1,R_2)-\frac{z}{32n^2}\right)\right)$$

and we estimate

$$\begin{aligned}&\#\left(\mathbb{Z}^3\cap\left(\frac{1}{32n^2}\cdot\mathcal{R}(N,R_1,R_2)-\frac{z}{32n^2}\right)\right)\\ =&\#\left\{(a-z_1/32n^2,b-z_2/32n^2,c-z_3/32n^2)\in\mathbb{Z}^3_{\geq 1}\mid ab^{2/3}c<\frac{N}{(32n^2)^{8/3}},\ a/c\in[1,R_1]\ \text{ and } b\in\left[\frac{-1}{32n^2},\frac{R_2}{32n^2}\right]\right\}\\ =&\sum_{b'=0}^{\lfloor(R_2-z_2)/32n^2\rfloor}\#\left\{(a-z_1/32n^2,c-z_3/32n^2)\in\mathbb{Z}^2_{\geq 1}\mid ac<\frac{N}{32^2n^4(32n^2b'+z_2)^{2/3}},\ a/c\in[1,R_1]\right\}\\ =&\sum_{b'=0}^{\lfloor(R_2-z_2)/32^2n^4\rfloor}\#\left(\mathcal{S}\left(\frac{N}{32^2n^4(32n^2b'+z_2)^{2/3}},R_1\right)-\left(\frac{z_1}{32n^2},\frac{z_3}{32n^2}\right)\right)\cap\mathbb{Z}^2\\ =&\sum_{b'=0}^{\lfloor(R_2-z_2)/32n^2\rfloor}\text{Area}\left(\mathcal{S}\left(\frac{N}{32^2n^4(32n^2b'+z_2)^{2/3}},R_1\right)\right)+O\left(\sum_{b'=0}^{\lfloor(R_2+z_2)/32n^2\rfloor}L\left(\frac{N}{32^2n^4(32n^2b'+z_2)^{2/3}},R_1\right)\right),\end{aligned}$$

where in the above chain of equalities, we set $b' := b - z_2/32n^2$. Note that in the sum, b does not take integer values, but b' does. From Lemma 4.2, we find that the area of $\mathcal{S}(M,R)$ is given by:

$$\text{Area}\left(\mathcal{S}(M,R)\right) = \begin{cases}\frac{M}{2}\log R - \frac{(R-1)}{2} & \text{if } M\geq R;\\ \frac{M}{2}\log M - \frac{(M-1)}{2} & \text{if } M<R.\end{cases}$$

Since $N > n^4 R_1 R_2$, we have that

$$\begin{aligned}
&\#\left(\mathbb{Z}^3 \cap \left(\frac{1}{32n^2}\cdot \mathcal{R}(N,R_1,R_2) - \frac{z}{32n^2}\right)\right)\\
=&\frac{N\log R_1}{2}\frac{1}{32^2 n^4}\left\{\sum_{\substack{b=1,\\ b\equiv z_2 \mod 32n^2}}^{\lfloor R_2\rfloor} b^{-2/3}\right\} + O_{n,R_1,R_2}(\sqrt{N})
\end{aligned}$$

Thus we find that

$$\begin{aligned}
\#\mathcal{C}_n^{\tau}(N,R_1,R_2) =&\frac{N\log R_1}{2}\frac{1}{32^2 n^4}\sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor}\frac{\#\{z\in\widetilde{\Omega}_n \mid z_2 = b \pmod{32n^2}\}}{b^{2/3}} + O_{n,R_1,R_2}(\sqrt{N})\\
=&\frac{N\log R_1}{2\times 32^2}\frac{1}{n^4}\sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor}\frac{n_\tau(b)\times\prod_{\ell\mid b}(\ell^2-\ell)^2\times\prod_{\ell\mid n,\ell\nmid b}\left(\ell^4-3\ell^2+2\ell\right)}{b^{2/3}} + O_{n,R_1,R_2}(\sqrt{N})\\
=&\frac{N}{2\times 32^2}\prod_{\ell\mid n}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1\sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor} n_\tau(b) b^{-2/3}\prod_{\ell\mid b}\left(\frac{\ell}{\ell+2}\right) + O_{n,R_1,R_2}(\sqrt{N}).
\end{aligned}$$

□

Let $\mathcal{C}'_\ell(N,R_1,R_2)$ be the complement of $\mathcal{C}_\ell(N,R_1,R_2)$ in $\mathcal{R}_{\mathbb{Z}^3}(N,R_1,R_2)$.

Lemma 4.9. *We have that*

$$\frac{\#\mathcal{C}'_\ell(N,R_1,R_2)}{N} = O\left(\frac{1}{\ell^{4/3}}+\frac{1}{\sqrt{N}}\right),$$

where the implied constant depends only on R_1 and R_2, and not on N or ℓ.

Proof. Suppose $(a,b,c)\in\mathcal{C}'_\ell(N,R_1,R_2)$, then there are the following cases to consider:

(1) $\ell^2|a$,
(2) $\ell^2|b$,
(3) $\ell^2|c$,
(4) $\ell|a$ and $\ell|b$,
(5) $\ell|a$ and $\ell|c$,
(6) $\ell|b$ and $\ell|c$.

For $i=1,\dots,6$, let $\mathcal{F}_i(N,R_1,R_2)$ be the subset of $\mathcal{C}'_\ell(N,R_1,R_2)$ consisting of (a,b,c) satisfying (i) above. First we bound $\#\mathcal{F}_1(N,R_1,R_2)$ we write $a=\ell^2 a'$. Note that (a',b,c) lies in the set

$$\{(a',b,c)\in\mathbb{Z}^3_{\geq 1} \mid a'b^{2/3}c\leq N/\ell^2,\quad a'/c\in[1/\ell^2,R_1/\ell^2],\quad b\in[1,R_2]\}.$$

We find that

$$\begin{aligned}
\#\mathcal{F}_1(N,R_1,R_2) =& \sum_{b=1}^{\lfloor R_2\rfloor}\#\{(a',c)\in\mathbb{Z}^2_{\geq 1} \mid a'c\leq\frac{N}{\ell^2 b^{2/3}},\quad a'/c\in[1/\ell^2,R_1/\ell^2]\}\\
=&\sum_{b=1}^{\lfloor R_2\rfloor} O\left(\text{Area}\left(\{(a',c)\in\mathbb{Z}^2_{\geq 1} \mid a'c\leq\frac{N}{\ell^2 b^{2/3}},\quad a'/c\in[1/\ell^2,R_1/\ell^2]\}\right)\right)\\
&+\sum_{b=1}^{\lfloor R_2\rfloor} O\left(L\left(\{(a',c)\in\mathbb{Z}^2_{\geq 1} \mid a'c\leq\frac{N}{\ell^2 b^{2/3}},\quad a'/c\in[1/\ell^2,R_1/\ell^2]\}\right)\right)\\
=&O_{R_1,R_2}(N/\ell^2),
\end{aligned}$$

where the area and length calculation is similar to that in Lemmas 4.2 and 4.3. We find that

$$\begin{aligned}
\#\mathcal{F}_2(N,R_1,R_2) &= \sum_{b'=1}^{\lfloor R_2/\ell^2 \rfloor} \#\{(a,c)\in\mathbb{Z}^2_{\geq 1} \mid ac \leq \frac{N}{\ell^{4/3}b'^{2/3}}, \quad a/c\in[1,R_1]\} \\
&= \sum_{b'=1}^{\lfloor R_2/\ell^2 \rfloor} O\left(\text{Area}\left(\{(a,c)\in\mathbb{Z}^2_{\geq 1} \mid ac \leq \frac{N}{\ell^{4/3}b'^{2/3}}, \quad a/c\in[1,R_1]\}\right)\right) \\
&+ \sum_{b'=1}^{\lfloor R_2/\ell^2 \rfloor} O\left(L\left(\{(a,c)\in\mathbb{Z}^2_{\geq 1} \mid ac \leq \frac{N}{\ell^{4/3}b'^{2/3}}, \quad a/c\in[1,R_1]\}\right)\right) \\
&= O_{R_1,R_2}\left(\frac{N}{\ell^{4/3}}\right) + O_{R_1,R_2}(\sqrt{N}), \\
\#\mathcal{F}_3(N,R_1,R_2) &= \sum_{b=1}^{\lfloor R_2 \rfloor} \#\{(a,c')\in\mathbb{Z}^2_{\geq 1} \mid ac' \leq N/\ell^2 b, \quad a/c'\in[1,\ell^2R_1]\} \\
&= O_{R_1,R_2}(N\log\ell/\ell^2) + O_{R_1,R_2}(\sqrt{N}), \\
\#\mathcal{F}_4(N,R_1,R_2) &= \sum_{b'=1}^{\lfloor R_2/\ell \rfloor} \#\{(a',c)\in\mathbb{Z}^2_{\geq 1} \mid a'c \leq N/\ell^{5/3}b', \quad a'/c\in[1,R_1/\ell]\} \\
&= O_{R_1,R_2}(N/\ell^{5/3}) + O_{R_1,R_2}(\sqrt{N}), \\
\#\mathcal{F}_5(N,R_1,R_2) &= \sum_{b=1}^{\lfloor R_2 \rfloor} \#\{(a',c')\in\mathbb{Z}^2_{\geq 1} \mid a'c' \leq N/\ell^2 b, \quad a'/c'\in[1,R_1]\} \\
&= O_{R_1,R_2}(N/\ell^2) + O_{R_1,R_2}(\sqrt{N}), \\
\#\mathcal{F}_6(N,R_1,R_2) &= \sum_{b'=1}^{\lfloor R_2/\ell \rfloor} \#\{(a,c')\in\mathbb{Z}^2_{\geq 1} \mid ac' \leq N/\ell^{5/3}b', \quad a/c'\in[1,\ell R_1]\} \\
&= O_{R_1,R_2}\left(N\log\ell/\ell^{5/3}\right) + O_{R_1,R_2}(\sqrt{N}).
\end{aligned}$$

Thus putting it all together, one finds that

$$\frac{\#\mathcal{C}'_\ell(N,R_1,R_2)}{N} = \sum_{i=1}^{6}\frac{\mathcal{F}_i(N,R_1,R_2)}{N} = O\left(\ell^{-4/3}+\frac{1}{\sqrt{N}}\right).$$

□

Lemma 4.10. *Let $Y>0$ be a positive constant and let $\mathcal{D}_Y(N,R_1,R_2) := \bigcup_{\ell>Y}\mathcal{C}'_\ell(N,R_1,R_2)$. Then we find that*

$$\frac{\#\mathcal{D}_Y(N,R_1,R_2)}{N} = O\left(\sum_{\ell>Y}\ell^{-4/3}+\frac{1}{\log N}\right).$$

Proof. Clearly $\mathcal{C}'_\ell(N,R_1,R_2)=\emptyset$ unless $\ell \ll \sqrt{N}$. Thus from Lemma 4.9,

$$\frac{\#\mathcal{D}_Y(N,R_1,R_2)}{N} = O\left(\sum_{\ell>Y}\ell^{-4/3}+\frac{\pi(\sqrt{N})}{\sqrt{N}}\right).$$

The result follows from the prime number theorem. □

Proposition 4.11. *We have that*

$$\begin{aligned}\lim_{N\to\infty}\frac{\#\mathcal{C}^\tau(N,R_1,R_2)}{N} =& \frac{1}{2048}\prod_{\ell}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1 \sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor} n_\tau(b)b^{-2/3}\prod_{\ell\mid b}\left(\frac{\ell}{\ell+2}\right),\\ =& \frac{1}{2048}\prod_{\ell}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1\psi_\tau(R_2),\end{aligned}$$

where

$$\psi_\tau(x) := \sum_{n=1}^{\lfloor x\rfloor} n_\tau(n)\alpha(n).$$

Proof. Let n be a squarefree natural number which is divisible by all primes $\ell \le R_2$. Then for $N > n^4R_1R_2$, Proposition 4.8 asserts that

$$\#\mathcal{C}_n^\tau(N,R_1,R_2) = \frac{N}{2048}\prod_{\ell\mid n}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1 \sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor} n_\tau(b)b^{-2/3}\prod_{\ell\mid b}\left(\frac{\ell}{\ell+2}\right)+O_{n,R_1,R_2}(\sqrt{N}).$$

For $Y>0$, set $n_Y := \prod_{\ell\le Y}\ell$. We find that

$$\limsup_{N\to\infty}\frac{\#\mathcal{C}^\tau(N,R_1,R_2)}{N} \le \lim_{N\to\infty}\frac{\#\mathcal{C}_{n_Y}^\tau(N,R_1,R_2)}{N} = \frac{1}{2048}\prod_{\ell\le Y}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1 \sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor} n_\tau(b)b^{-2/3}\prod_{\ell\mid b}\left(\frac{\ell}{\ell+2}\right).$$

Taking the limit of the right hand side as $Y\to\infty$, we find that

$$\limsup_{N\to\infty}\frac{\#\mathcal{C}^\tau(N,R_1,R_2)}{N} \le \frac{1}{2048}\prod_{\ell}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1 \sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor} n_\tau(b)b^{-2/3}\prod_{\ell\mid b}\left(\frac{\ell}{\ell+2}\right).$$

On the other hand, since

$$\mathcal{C}_{n_Y}^\tau(N,R_1,R_2) \subseteq \mathcal{C}^\tau(N,R_1,R_2)\cup\mathcal{D}_Y(N,R_1,R_2),$$

we find that

$$\begin{aligned}\liminf_{N\to\infty}\frac{\#\mathcal{C}^\tau(N,R_1,R_2)}{N} &\ge \lim_{N\to\infty}\frac{\#\mathcal{C}_{n_Y}(N,R_1,R_2)}{N} - \limsup_{N\to\infty}\frac{\#\mathcal{D}_Y(N,R_1,R_2)}{N}\\ &\ge \frac{1}{2048}\prod_{\ell\le Y}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1 \sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor} n_\tau(b)b^{-2/3}\prod_{\ell\mid b}\left(\frac{\ell}{\ell+2}\right) - C\sum_{\ell>Y}\ell^{-4/3}\end{aligned}$$

for some positive constant $C>0$. Taking the limit of the right hand side with $Y\to\infty$, we find that

$$\liminf_{N\to\infty}\frac{\#\mathcal{C}^\tau(N,R_1,R_2)}{N} \ge \frac{1}{2048}\prod_{\ell}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1 \sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor} n_\tau(b)b^{-2/3}\prod_{\ell\mid b}\left(\frac{\ell}{\ell+2}\right).$$

Therefore, we have shown that

$$\lim_{N\to\infty}\frac{\#\mathcal{C}^\tau(N,R_1,R_2)}{N} = \frac{1}{2048}\prod_{\ell}\left(1-\frac{3}{\ell^2}+\frac{2}{\ell^3}\right)\log R_1 \sum_{\substack{b=1,\\ b\,\square\text{-free}}}^{\lfloor R_2\rfloor} n_\tau(b)b^{-2/3}\prod_{\ell\mid b}\left(\frac{\ell}{\ell+2}\right).$$

□

We now give a proof of the main result.

Proof of Theorem 1.4. Assume that $m > 0$ and is of Type II. The argument in the remaining cases proceeds similarly. Write $m = ab^2c^3$, where (a, b, c) is a carefree triple with $a, b, c > 0$, and suppose further that $c/a > 1$. This condition ensures that the isomorphism class of the number field K_m is uniquely determined. Recall from Corollary 3.2 that the discriminant of K_m is given by $\Delta_m = -2^8a^3b^2c^3$. Setting $X = 2^8N^3$, the condition $|\Delta_m| \leq X$ becomes equivalent to

$$ab^{2/3}c \leq N.$$

Consider the four rectangles:

$$\mathcal{R}_1 := [1, R_1] \times [1, R_2],$$
$$\mathcal{R}_2 := [1, R_1] \times [1, R_2'],$$
$$\mathcal{R}_3 := [1, R_1'] \times [1, R_2],$$
$$\mathcal{R}_4 := [1, R_1'] \times [1, R_2'],$$

so that by inclusion–exclusion,

$$\int_{\mathcal{R}} = \int_{\mathcal{R}_1} - \int_{\mathcal{R}_2} - \int_{\mathcal{R}_3} + \int_{\mathcal{R}_4}.$$

It therefore suffices to prove the theorem under the simplifying assumption $R_1' = R_2' = 1$, in which case the claim follows directly from Proposition 4.11. □

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